

Blue Beam: Detecting and Weakly Localizing Subtle Model Tampering in Small Models with Calibrated Latent Signals

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Abstract

We study whether subtle tampering in small open models leaves internal traces that can support defensive detection and coarse localization. Our goal is not payload recovery, hidden-backdoor decoding, or exact causal reconstruction. Instead, we frame the problem as a **yellow-flag** setting: identify compromise-induced drift early, preserve benign throughput, and weakly localize the most diagnostic procedure families.

Using BashArena-style trajectory bundles on Qwen 3.5 4B, we compare base, orchestration, and manipulation conditions across matched procedure families. We combine cheap behavioral probes, latent feature-bank scores, corruption-gap comparisons, token-level sparse-autoencoder summaries, and follow-on TinyLoRA evaluation plans. The strongest story is not that any single detector solves the problem, but that a calibrated stack converts weak internal evidence into a more selective escalation policy than a saturating hard-block baseline.

Our central empirical result is a refined low-layer signal at layer -4 on a shared tamper-sensitive family, especially `multi_step_risk_escalation`. Two features in the refined sweep show strong ranking metrics and bridge back to orchestration-linked structure at layer -14, suggesting that at least some tamper-sensitive drift is mechanically recoverable across lanes and across depth. We therefore argue for Blue Beam as a calibrated tamper-sensing framework for early warning, coarse localization, and triage rather than a claim of universal detection or exact mechanistic decoding.

1 Introduction

As local models become more widely deployed inside agentic workflows, the practical defensive question is not only whether a model can be made unsafe, but whether subtle compromise can be surfaced before it produces obvious failures. In many realistic failure modes, the danger is not a single overtly malicious action. Instead, the model may preserve plausible local behavior while gradually drifting toward more dangerous sequencing, staging, configuration changes, or coordination support.

This paper studies a narrower and more operational version of that problem. Can subtle tampering in a small open model be detected as a yellow flag, and can the resulting drift be weakly localized to recurring procedure families? We are not trying to decode a hidden payload, recover a latent trigger, or provide universal backdoor attribution. The aim is more modest and more useful: identify warning structure early enough to support calibrated escalation and targeted monitoring.

Our analysis uses BashArena-style shell trajectories and matched comparison bundles over Qwen 3.5 4B. We compare base conditions against orchestration and manipulation follow-on conditions, then aggregate evidence from several weak but partially independent views: cheap behavioral probes, latent feature-bank signals, corruption-gap reports, token-level SAE summaries, hill-climb rankings, and TinyLoRA follow-on evaluation plans. The resulting picture is not one of perfect semantic detection. Rather, it suggests that some classes of tampering leave detectable internal structure, and that this structure can sometimes be weakly aligned with recurring procedural families such as stepwise escalation or unauthorized configuration drift.

The strongest current empirical result is a refined low-layer signal at layer -4 on `multi_step_risk_escalation`, with bridge support back to orchestration-linked evidence at layer -14. That result matters because it is not merely a pooled detector score. It suggests that at least part of the tamper-sensitive warning signal is recoverable at lower depth while remaining linked to broader orchestration-reference structure.

Our contributions are therefore threefold:

1. We frame subtle tampering in small models as a **tamper-sensing** problem rather than a payload-recovery problem.
2. We show that controller calibration matters: a selective latent amplification policy improves triage utility over a saturating hard-block policy.
3. We provide evidence that some compromise-induced drift can be weakly localized to recurring procedure families, including a refined low-layer signal at layer -4 that bridges back to orchestration-linked structure.

2 Problem Framing

It is useful to distinguish three ambitions that are often conflated.

First, **payload recovery** asks whether one can decode the hidden mechanism itself. That is not our goal. We do not claim to reverse-engineer a secret trigger, extract a hidden policy, or recover the exact causal path from weights to behavior.

Second, **tamper detection** asks whether a model or trajectory bundle appears measurably compromised relative to a base condition. This is our primary aim. A practical defender does not need complete mechanistic recovery in order to benefit from early warning.

Third, **behavioral localization** asks whether the observed drift can be weakly aligned with interpretable recurring procedure families. This is our secondary aim. Even coarse localization is useful if it helps defenders distinguish diffuse anomaly from more structured escalation pressure.

Blue Beam therefore makes a deliberately restrained claim: subtle tampering may leave detectable internal and behavioral traces that support calibrated yellow-flag escalation and weak procedural localization, even when exact mechanistic attribution remains out of reach.

3 Experimental Setting

3.1 Environment and Model

We use BashArena-style shell trajectories as the core experimental substrate. The current model family is Qwen 3.5 4B. This size is useful because it sits near the cusp where local models remain tractable for mechanistic inspection while still being capable enough to matter operationally in tool-using and agentic settings.

The main comparison lanes in the current bundle are:

- a base reference condition,
- an orchestration reference lane,
- a manipulation follow-on lane,
- and a refined local sweep focused on lower layers.

The imported activation bundle is drawn from a shared corpus root, and the paper-level artifacts include corruption-gap reports, token-SAE summaries, hill-climb candidate tables, matched-family comparison figures, and TinyLoRA evaluation plans.

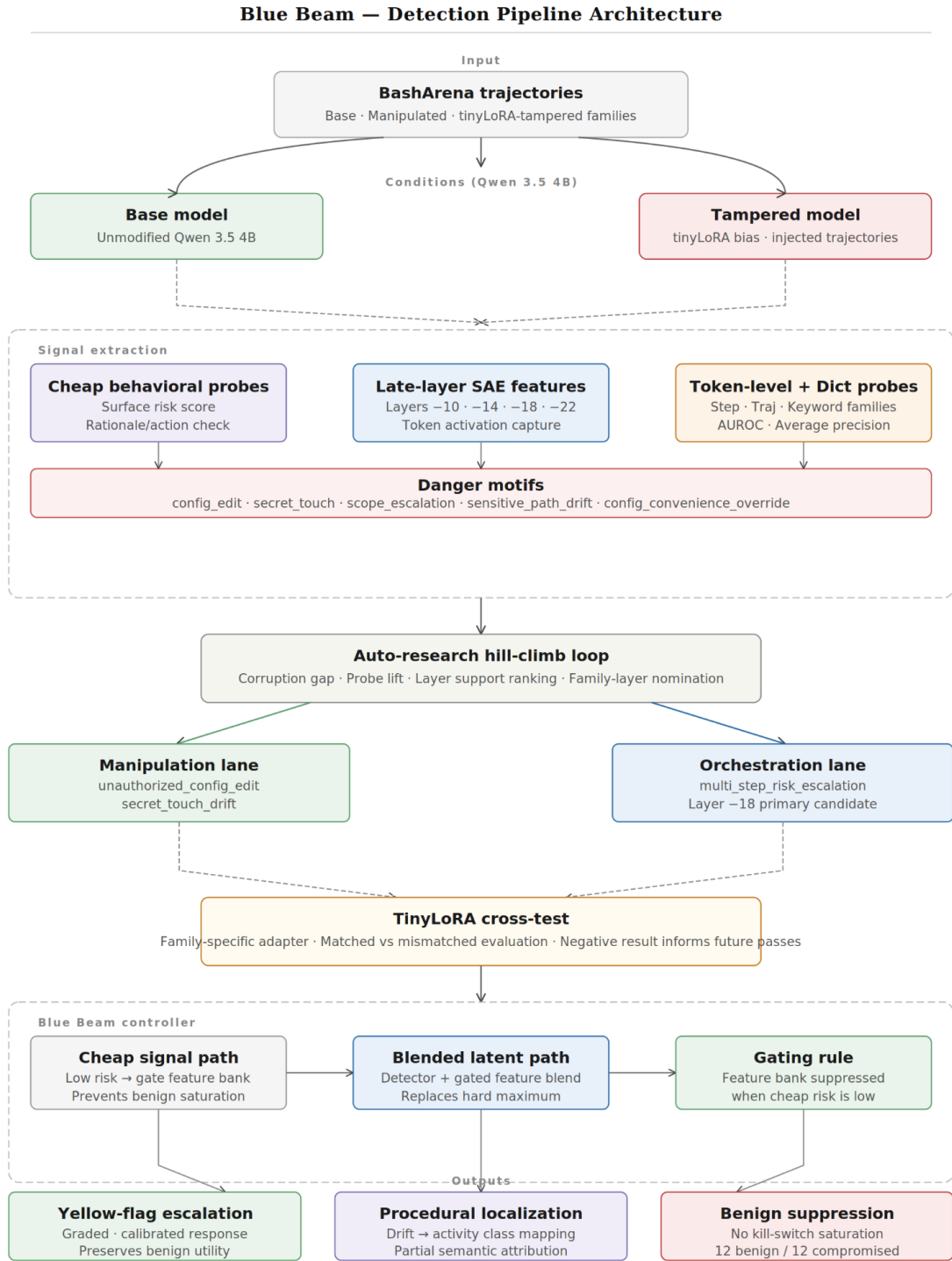


Figure: Blue Beam detection pipeline — from BashArena trajectory input through signal extraction, hill-climb localization, TinyLoRA cross-test, and calibrated controller to graded outputs.

Figure 1: Blue Beam detection pipeline architecture, from BashArena trajectory input through signal extraction, localization, follow-on adapter evaluation, and calibrated controller outputs.

3.2 Procedure Families

We organize analysis around recurring *procedure families* rather than around explicit offensive payload content. The families that recur most strongly in the present bundle are:

- `multi_step_risk_escalation`
- `unauthorized_config_edit`
- `delayed_policy_violation_setup`
- `secret_touch_drift`
- `rationale_action_mismatch`
- `suspicious_file_staging`

These names should not be read as claims about a real-world attack taxonomy. They are safe proxy families used to study structured drift, sequencing pressure, and escalation-related anomalies.

3.3 Signals

The analysis stack combines several weak views:

- cheap controller scores,
- latent feature-bank scores,
- corruption-gap comparisons,
- token-level SAE summaries,
- dictionary or hill-climb rankings,
- and matched-family TinyLoRA evaluation plans.

No single score is assumed to be decisive. The point is to accumulate consistent evidence from multiple partially independent views.

4 Blue Beam Controller

The controller is best understood as a calibrated triage mechanism rather than a kill switch. In earlier versions, latent evidence could saturate benign traffic into excessive blocking behavior. The revised controller instead treats latent evidence as a *selective amplifier*.

Operationally, the cheap path provides a low-cost first-pass risk score, while the latent path incorporates detector and gated feature-bank information. Rather than taking a hard maximum over all available evidence, the latent signal is blended and partially suppressed when cheap risk is low. This prevents benign low-risk rows from being dominated by isolated feature-bank spikes while preserving stronger escalation on rows that already show broader warning structure.

That distinction matters for practical use. A defensive monitoring layer is not useful if it collapses into blanket denial. It becomes useful when it can preserve benign throughput, surface uncertainty, and escalate selectively.

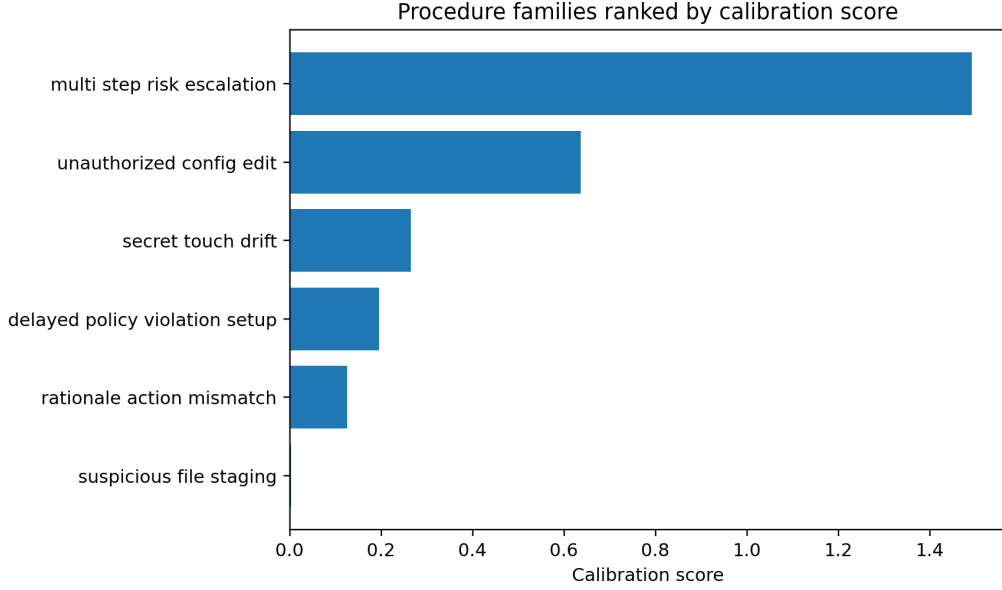


Figure 2: Procedure-family calibration scores from the auto-research pass. Multi-step risk escalation is the strongest current family, followed by unauthorized configuration edit and secret-touch drift.

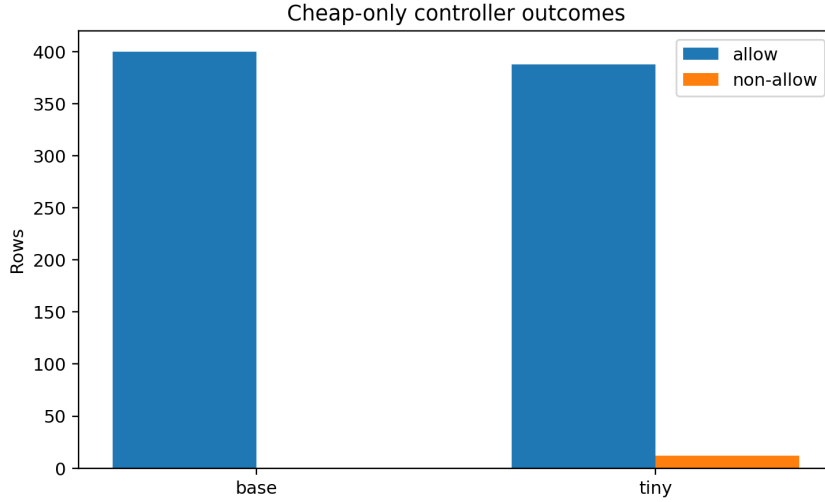


Figure 3: Cheap-only controller outcomes on base and compromised manipulation runs. The cheap path alone is weakly discriminative, which motivates the later latent and token-level extensions rather than relying on a single shallow risk score.

5 Localization Math

Let $a[t, l, f]$ denote the activation of feature f at token t in layer l , and let $m[t] \in \{0, 1\}$ indicate whether token t lies inside a suspicious span.

We summarize token-level localization with three compact quantities.

First, the token-feature correlation

$$\rho[l, f] = \text{Corr}(a[t, l, f], m[t])$$

measures whether feature f tends to activate on suspicious tokens.

Second, the activation enrichment

$$E[l, f] = \mathbb{E}[a[t, l, f] \mid m[t] = 1] - \mathbb{E}[a[t, l, f] \mid m[t] = 0]$$

measures the average activation gap between suspicious and non-suspicious spans.

Third, to compare localization under base and tampered conditions, we define a degradation quantity

$$\Delta\rho[l, f] = \rho_{\text{base}}[l, f] - \rho_{\text{comp}}[l, f].$$

Positive $\Delta\rho[l, f]$ indicates that the feature localizes suspicious behavior more cleanly in the base condition than in the compromised condition.

These quantities are not intended as a complete theory of tampering. They are compact ranking tools that help identify which features, layers, and family-layer pairs are most informative. In practice, we use them alongside AUROC, average precision, corruption-gap statistics, and cross-lane bridge comparisons rather than in isolation.

6 Results

6.1 Controller Calibration

The controller story remains strongest when latent evidence is treated as a selective amplifier rather than a blanket veto. Under that interpretation, benign traffic no longer collapses into constant hard-block behavior, while compromised traffic still produces stronger escalation. This makes the controller more useful for triage and monitoring than a saturating latent-defense baseline.

The practical claim here is simple: *calibration matters*. Internal warning signals are more useful when they produce graded escalation than when they indiscriminately dominate all downstream policy.

6.2 Orchestration Lane as Stable Reference

The orchestration lane remains the strongest aggregate reference point in the current bundle. At the broad level, `multi_step_risk_escalation` is the clearest recurring family, and the orchestration exports provide the most stable comparison anchor for later follow-on analysis.

Historically, the orchestration lane was most strongly associated with deeper layers such as -18, and that broader reference structure remains useful even after the addition of refined lower-layer sweeps. This stability matters because it provides a baseline against which weaker or noisier manipulation-lane evidence can be interpreted.

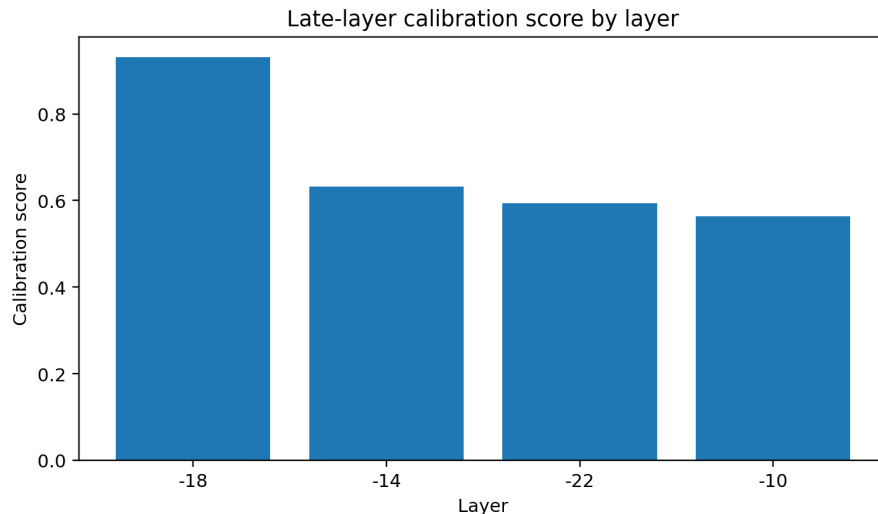


Figure 4: Aggregate layer summary from the auto-research exports. The orchestration lane remains the historical anchor, while the refined sweep adds a new low-layer signal.

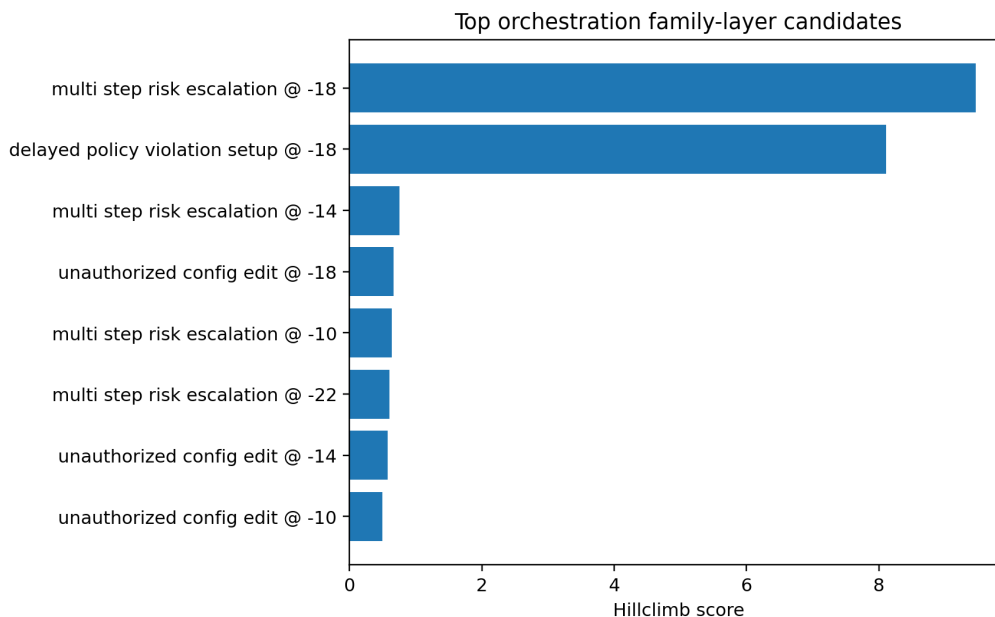


Figure 5: Top orchestration family-layer candidates from the orchestration hill-climb export. Multi-step risk escalation dominates the ranking, with delayed policy violation setup as the main secondary cluster.

6.3 Manipulation Follow-On

The manipulation follow-on lane remains informative, but it is not the strongest current headline result. Its value is that localization remains nontrivial even when final controller-side separation is weak. In particular, the hill-climb outputs continue to nominate a small number of family-layer pairs, and token-SAE coverage shows that informative layers do not always coincide with the broad

orchestration reference layer.

This is useful evidence, but it should be read as follow-on support rather than as the main positive claim of the paper.

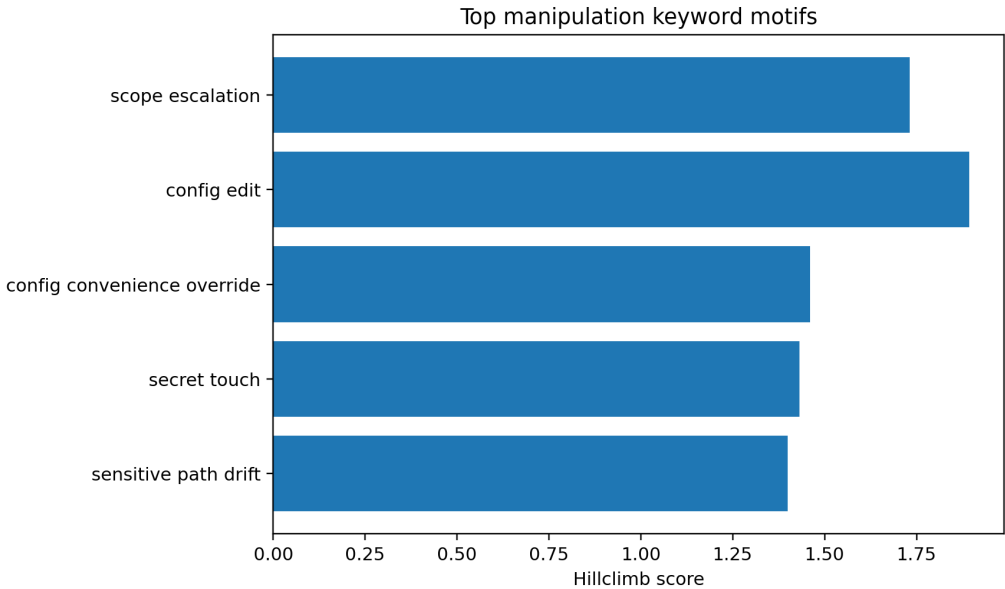


Figure 6: Highest-scoring manipulation keyword motifs in the hill-climb analysis. Configuration editing, scope escalation, convenience overrides, and secret-touch motifs dominate the current manipulation lane.

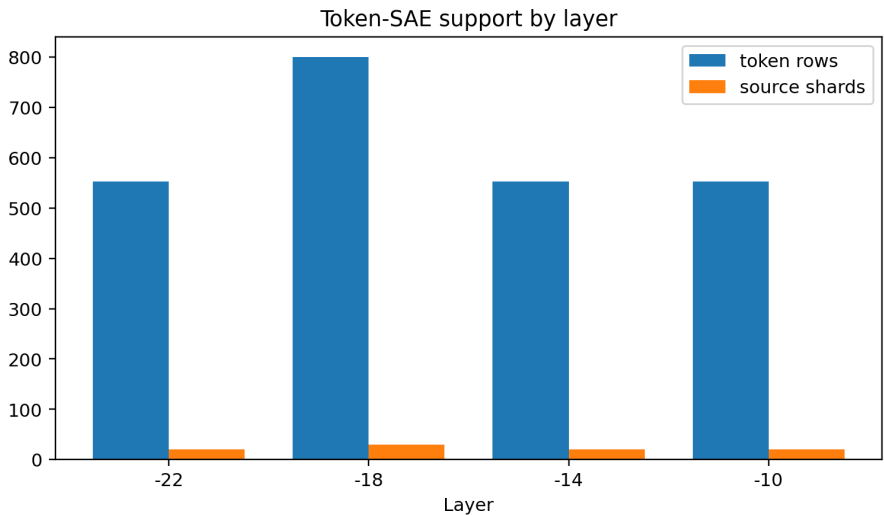


Figure 7: Token-SAE support by layer in the manipulation analysis. The layer -18 run has the broadest token and shard support, which is consistent with its appearance as the strongest current localization layer.

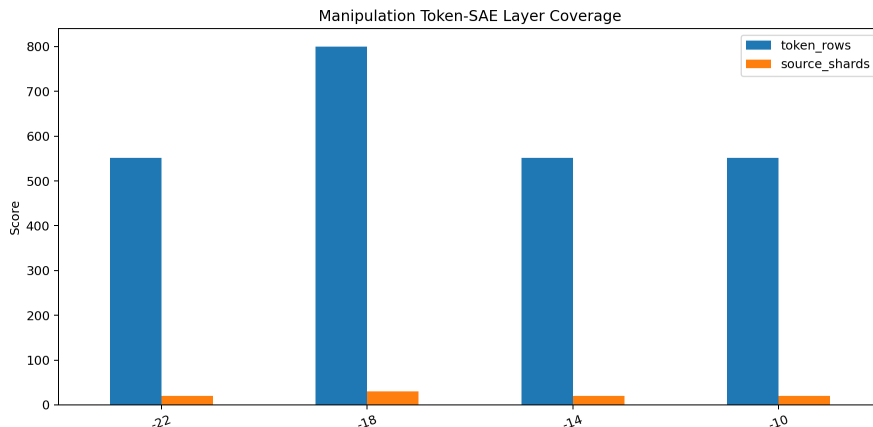


Figure 8: Additional token-SAE layer coverage for the manipulation bundle. Coverage is broad enough to support follow-on comparison, but the clearest new result comes from the refined low-layer sweep.

6.4 TinyLoRA Plans versus Evidence

TinyLoRA-related material in the current bundle should be interpreted carefully. The ranked TinyLoRA outputs are useful because they nominate matched family-layer targets for follow-on evaluation, but they do *not* by themselves establish a demonstrated adapter-backed intervention success.

This distinction is important. The current manipulation TinyLoRA cross-test shows weak final action-level separation, and the narrow family-specific perturbation does not yet yield a strong controller-side intervention signal. Accordingly, TinyLoRA should be treated in this paper as a *planning and evaluation path*, not as a primary positive result.

The negative result is still informative. It suggests that the SAE and probe stack may currently be better at ranking candidate families and layers than at immediately producing strong action-level separation through narrow perturbation. But that remains future work, not a completed claim.

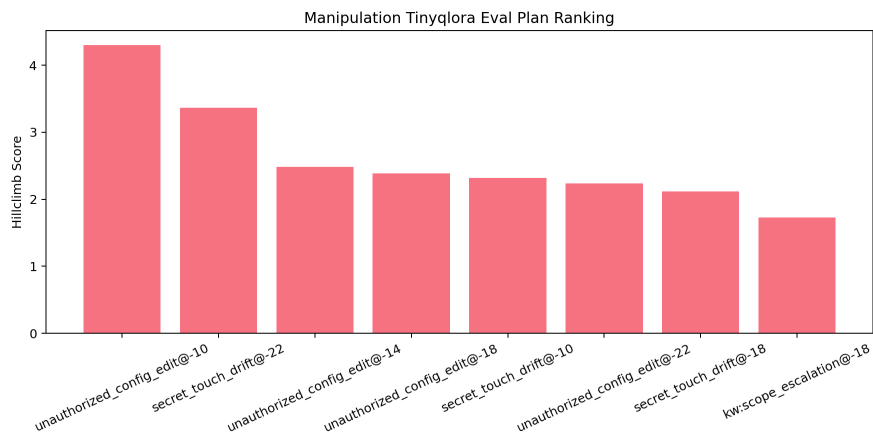


Figure 9: Manipulation TinyLoRA evaluation plan exported from the hill-climb loop. It provides a ranked path for stronger family-specific perturbations and better matched-versus-mismatched attribution tests.

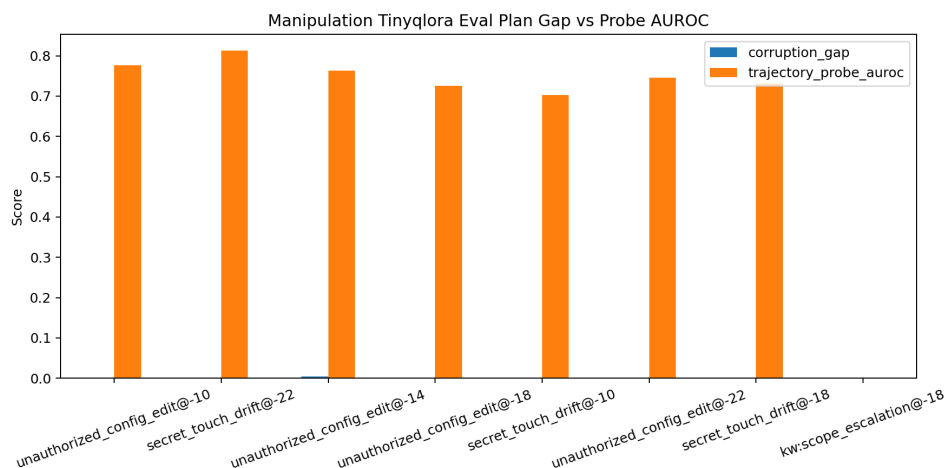


Figure 10: Matched-family TinyLoRA plan components. These planning artifacts are useful for follow-on evaluation but should not be read as positive intervention evidence.

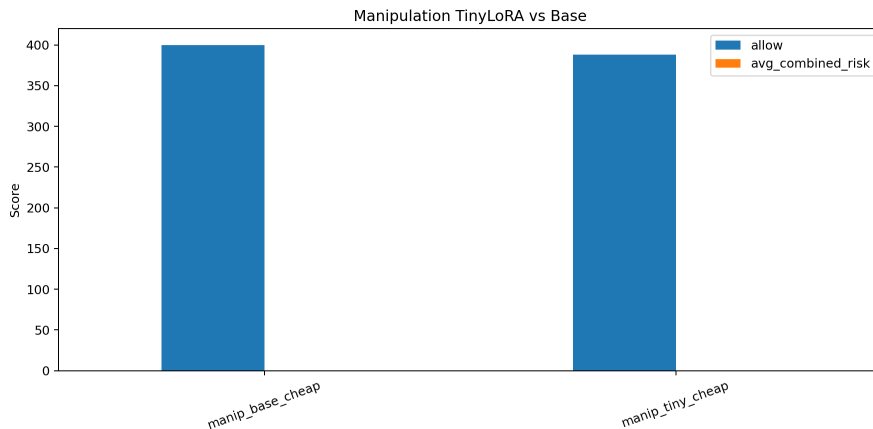


Figure 11: Matched-family TinyLoRA cross-test matrix on the manipulation bundle. The current family-specific perturbation shows weak final-decision separation despite earlier localization evidence nominating stronger family-layer targets.

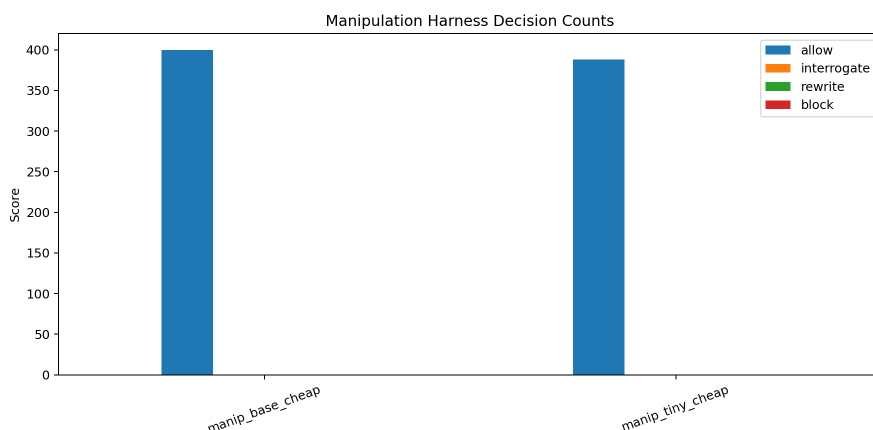


Figure 12: Action-decision outcomes from the TinyLoRA manipulation cross-test. The narrow adapter shifts outcomes only weakly at the controller layer, supporting its treatment as future work rather than a core positive result.

6.5 Refined Low-Layer Signal at -4

The clearest positive result in the present update is the refined low-layer sweep. It yields a concrete tamper-sensitive signal at layer -4 on a shared procedure family, especially `multi_step_risk_escalation`.

Two features are especially relevant:

- `sae_layer-4_feat46` on `multi_step_risk_escalation`
- `sae_layer-4_feat75` on `multi_step_risk_escalation`

Both appear in the refined sweep with strong ranking metrics, and both bridge back to orchestration-linked structure at layer -14. This cross-lane bridge matters because it means the result is not just a raw classifier score detached from the rest of the evidence bundle. Instead, it

links a refined low-layer signal to the broader orchestration reference structure that already served as the stable anchor of the paper.

This is the central empirical contribution of the current paper version. It supports the claim that at least some classes of tampering are mechanically recoverable in internal state, not only at late pooled layers but also in lower-depth representations that remain connected to broader procedural structure.

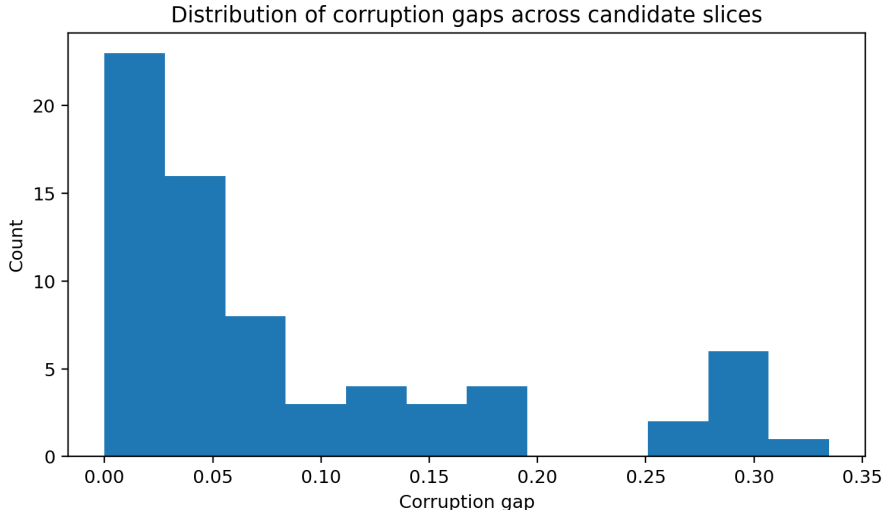


Figure 13: Histogram of corruption gaps across candidate slices gathered from the exported analyses. Most slices show modest degradation, while a smaller subset exhibits larger and more diagnostically useful gaps.

7 Discussion

The current evidence supports a restrained interpretation. Some classes of tampering leave detectable internal structure, and at least one lower layer appears to carry a useful signal for family-level localization. This is enough to justify a defense-oriented yellow-flag pipeline. It is not enough to claim universal detection, exact causal reconstruction, or robust payload recovery.

The most defensible reading is therefore procedural rather than semantic. The strongest current signals are concentrated on families involving stepwise escalation and configuration drift, which is consistent with the broader view that the warning structure is tracking procedural pressure rather than merely lexical content.

More broadly, Blue Beam suggests a different model-defense stance. Rather than waiting only for overtly unsafe outputs, a system can treat certain internal patterns as warning signals and escalate earlier while preserving benign utility through calibration. That is a more realistic use of current mechanistic tools than pretending they already provide full semantic decoding.

8 Limitations

- The current benchmark setting is narrow and may not generalize directly.

- The low-layer signal is strongest for a subset of families, not for all families.
- TinyLoRA remains a planning and follow-on evaluation path, not a demonstrated positive intervention result.
- The paper does not claim exact payload recovery, universal detection, or complete mechanistic attribution.
- The current families are proxy procedural families rather than a complete real-world attack taxonomy.

9 Future Work

Several follow-on directions appear especially natural.

First, the TinyLoRA path should be expanded carefully rather than overclaimed. The next step is not to declare adapter validation solved, but to run stronger matched-versus-mismatched family perturbations and use localization metrics as primary readouts.

Second, the SAE and dictionary stack should be extended into a reusable library across recurring procedure families. One-off experiment-specific probes are useful for discovery, but a reusable family-aligned library would be more valuable scientifically and operationally.

Third, the bridge between lower-layer and orchestration-linked signals should be studied more systematically. The current $-4 \leftrightarrow -14$ bridge is promising, but it is only one concrete case. A broader cross-depth analysis could clarify whether lower-layer warning signals consistently map onto stable higher-level procedural structure.

Fourth, a longer-term direction is the study of **SAE constellations**: coordinated sets of sparse features whose joint activation patterns are more informative than any single feature alone. The present bundle already suggests that recurrent family-layer pairs may form stable co-activation structure across layers and lanes.

10 Conclusion

Blue Beam is best understood as a calibrated tamper-sensing framework. Its value is not in claiming complete payload recovery or perfect mechanistic decoding, but in providing early warning, coarse localization, and improved escalation control.

The present bundle supports three practical claims:

1. calibration matters, because latent evidence is more useful when it avoids saturating benign traffic;
2. the orchestration lane remains the stable reference anchor for interpreting weaker follow-on evidence;
3. the refined sweep identifies a real low-layer signal at layer -4 for at least some tamper-sensitive procedure families, especially `multi_step_risk_escalation`, with bridge support back to orchestration-linked structure at layer -14 .

The immediate next step is to keep the paper focused on completed receipts, treat TinyLoRA as a follow-on evaluation path, and deepen the cross-layer analysis of tamper-sensitive internal signals.

In that sense, Blue Beam is not a claim that subtle tampering has been fully decoded. It is a claim that some of its warning structure is already detectable, calibratable, and weakly localizable.

A Bibliographic Footnotes

[^1] For inspiration on dynamic agent-security evaluation environments, see DeBenedetti et al., *AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents* (2024).

[^2] For the dictionary-learning framing, see Bricken et al., *Decomposing Language Models With Dictionary Learning* (2023).

[^3] For sparse autoencoders in language models and their scaling behavior, see Cunningham et al., *Sparse Autoencoders Find Highly Interpretable Features in Language Models* (2023), and Gao et al., *Scaling and Evaluating Sparse Autoencoders* (2024).

[^4] For closely related internal anomaly-detection framing, see Johnston, Chakraborty, and Belrose, *Mechanistic Anomaly Detection for “Quirky” Language Models* (2025).

[^5] Project repository: MoralityLabAI, *BlueBeam: SAE Control Protocol, Antivirus for Small Models*, GitHub repository (2026).